Tuple Perturbation-Based Contrastive Learning Framework for Multimodal Remote Sensing Image Semantic Segmentation

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***Abstract*— Deep learning models exhibit promising potential in multimodal remote sensing image semantic segmentation (MRSISS). However, the constrained access to labeled samples for training deep learning networks significantly influences the performance of these models. To address that, self-supervised learning (SSL) methods have garnered significant interest in the remote sensing community. Accordingly, this article proposes a novel multimodal contrastive learning framework based on tuple perturbation, which includes the pretraining and fine-tuning stages. First, a tuple perturbation-based multimodal contrastive learning network (TMCNet) is designed to better explore shared and different feature representations across modalities during the pretraining stage and the tuple perturbation module is introduced to improve the network's ability to extract multimodal features by generating more complex negative samples. In the fine-tuning stage, we develop a simple and effective multimodal semantic segmentation network (MSSNet), which can reduce noise by using complementary information from various modalities to integrate multimodal features more effectively, resulting in better semantic segmentation performance. Extensive experiments have been carried out on two published multimodal image datasets including optical and synthetic aperture radar (SAR) pairs, and the results show that the proposed framework can obtain more superior performance of semantic segmentation than the current state-of-the-art methods in cases of limited labeled samples. The source code is available at https://github.com/yeyuanxin110/TMCNet-MSSNet.**

***Index Terms*— Contrastive learning, multimodal remote sensing image (RSI), negative samples, semantic segmentation, tuple perturbation.**

Received 17 April 2024; revised 15 October 2024 and 2 January 2025; accepted 9 February 2025. Date of publication 20 February 2025; date of current version 5 March 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFC3015404, in part by the National Natural Science Foundation of China under Grant 42271446, and in part by the Central Guidance for Local Science and Technology Development Fund Projects of Sichuan Province under Grant 2024ZYD0084. *(Corresponding author: Liang Zhou.)*

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Digital Object Identifier 10.1109/TGRS.2025.3542868

I. Introduction

REMOTE sensing image semantic segmentation (RSISS) aims to accurately classify each pixel in an image into specific semantic categories, that is, to perform pixel-level image segmentation, so as to more accurately understand and describe different objects and scenes in the image [1]. Its main application areas include land cover classification [2], [3], [4], [5], cartography [6], [7], natural resource management, environmental monitoring [8], geospatial information analysis, and other fields. Therefore, RSISS is a crucial foundational task in the analysis of images [9], [10].

RSISS mainly includes semantic segmentation of optical, synthetic aperture radar (SAR), infrared, multispectral, hyperspectral, or other dataset [11]. Since it is difficult to break through the limitation of information diversity when only using the single-modal remote sensing image (RSI), the performance of semantic segmentation inevitably encounters bottlenecks in complex scenes. Compared with single-modal images, semantic segmentation of multimodal images can obtain a more comprehensive segmentation map of ground objects by making full use of complementary information between images [12]. Optical and SAR images represent two common forms of multimodal remote sensing data. The former offers detailed textures and superior resolution but struggles under adverse weather conditions, making it challenging to discern surface information during nighttime or cloud cover. Meanwhile, SAR, capable of all-weather, day-and-night imaging with cloud-penetrating abilities, primarily captures structural and outline features of surface objects, but it tends to be affected by the presence of speckle noise. Accordingly, the fusion of optical images and SAR images can effectively reduce the influence of poor weather on optical images and the serious speckle noise of SAR images [13], which has attracted much attention in multimodal RSISS (MRSISS). However, due to pronounced differences in imaging between optical and SAR images, it is quite different to fully tap and jointly utilize the information of these two types of multimodal data, making the traditional methods inevitably hit a bottleneck.

Currently, the explosion of remote sensing data makes it possible to obtain multimodal RSI within the same area, which offers a unique chance to improve semantic segmentation accuracy through leveraging the complementary characteristics of diverse imaging modes [14], [15]. Recently, supervised

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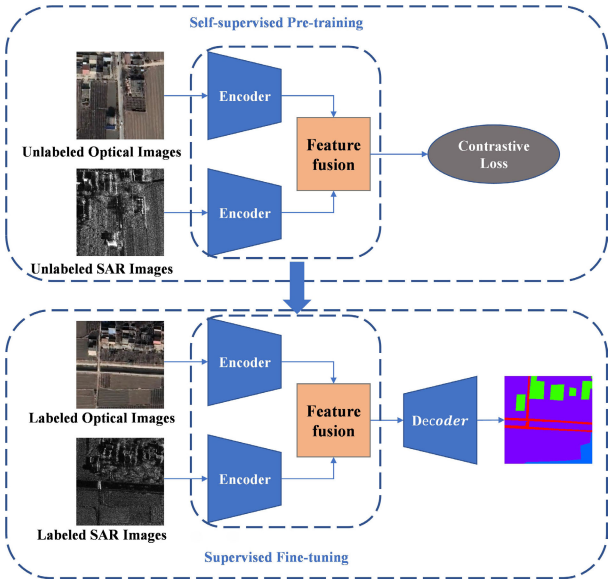


Fig. 1. Multimodal contrastive learning framework.

deep learning has a measure of success in MRSISS [16]. Hong et al. [17] first propose a universal and unified multimodal deep learning framework, providing an effective solution for semantic segmentation based on optical and SAR image fusion. Li et al. [18] present an MRSISS model named MCANet, which employs a multimodal cross-attention module to fuse the features of optical and SAR images based on the framework of DeepLabV3+ [19]. However, the MRSISS methods based on supervised deep learning rely heavily on a lot of high-quality labeled samples, which are very hard to get, time-consuming, and labor-intensive [20]. With the MRSISS technology playing an increasingly crucial role in rapid global development [21], [22], the issue of insufficient labeled samples is becoming increasingly severe.

To solve this problem, self-supervised learning (SSL) [23], [24], [25] offers an innovative framework. Initially, it acquires knowledge from unlabeled image data through the creation of self-supervised signals, subsequently applying this knowledge to downstream tasks. This approach reaches a performance level on these tasks that rivals that of supervised learning, even with a minimal number of labeled samples [26], [27], [28], [29], [30], [31]. As a typical SSL method, contrastive learning has made great achievements in the field of computer vision (CV). Chen et al. [23] present a contrastive learning framework of visual representations (SimCLR), which uses larger batch size, more data augmentation, the projection head, and other technologies, and achieve very good results on ImageNet. Recent research has integrated contrastive learning into remote sensing, successfully validating its applicability. To apply the contrastive learning method to the task of MRSISS, we propose a novel multimodal contrastive learning framework, as illustrated in Fig. 1. During the pretraining phase, the network conducts feature fusion on unlabeled optical and SAR images. The fused features are subsequently used as single-modal information for contrastive learning. Finally, we fine-tune the semantic segmentation network using a limited set of labeled images.

In addition, the existing contrastive learning mainly focuses on the learning of single-modal features and ignores the relationship between multimodal features [32]. There is a balance between learned optical feature and SAR feature in MRSISS tasks, and the primary challenge is how to efficiently extract and fuse multimodal features. To solve this problem, Liu et al. [33] present TupleInfoNCE, which is regarded as a new contrastive learning objective. The contrasting of sample pairs involves more than distinguishing between positive and negative correspondences; it also involves the creation of novel negative sample pairs, each depicted through various modalities to represent distinct scenes. Incorporating these supplemental negatives into training can promote the examination of intermodal correspondences within tuples by the learning model, safeguarding against the oversight of weak modalities. Inspired by it, we design a multimodal contrastive learning framework based on tuple perturbation, which overcomes the problem of feature interaction between different modalities in the feature extraction process of TupleInfoNCE. During the pretraining stage, we design a tuple perturbation-based multimodal contrastive learning network (TMCNet). The pseudo-Siamese feature extraction module is used to extract the features of optical and SAR images to avoid mutual interference of different modal features. Then, the tuple perturbation module is employed to better explore the shared and different feature representations between modalities and improve the network's ability to extract multimodal features by generating more complex negative samples. In addition, unlike CV, which pays more attention to the global features of images, RSISS requires a balance between global features and local features. For this reason, we adopt a loss function combining global contrastive loss and local contrastive loss to produce better feature representations. Moreover, in the fine-tuning stage, we design a simple and effective multimodal semantic segmentation network (MSSNet). In such a network, a multimodal feature rectification module (MFRM) is designed to reduce noise by using the complementary information between multimodalities, and a multiscale multimodal feature fusion module (MMFFM) is also designed to achieve an elegant collection of cross-attention and feature augmentation, fusing multimodal features more closely and improving semantic segmentation more effectively, including low-level feature fusion module (LFFM) and high-level feature fusion module (HFFM).

The primary contributions of this article are outlined as follows.

1) We apply the self-supervised contrastive learning to the MRSISS and propose a novel multimodal contrastive learning framework based on tuple perturbation. Validated across diverse datasets, the proposed framework allows the model to autonomously learn from unlabeled images and leverage a minimal set of annotations to direct the tasks involving multimodal semantic segmentation.
2) We propose a TMCNet. The TMCNet is proposed to better enhance the extraction of shared and different feature representations between patterns during the pretraining stage, improving the network's ability to extract

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multimodal features by generating more complex negative samples.

3) The MSSNet is proposed, which uses the complementary information between multimodal to reduce noise and fuse multimodal features more closely, thereby enhancing semantic segmentation results.

The rest of this article is structured as follows. Section II introduces the related work on contrastive learning and MRSISS. The methodology proposed is detailed in Section III. Section IV showcases a broad range of comparative analyses and an ablation study. Finally, Section V gives the conclusion.

II. Related Works

A. Contrastive Learning

SSL [20], [34], [35] originated from natural language processing (NLP). During the pretraining stage, the network learns data representation from unlabeled raw data, and then, in the fine-tuning stage, it uses a small quantity of labeled data for training to achieve enhanced network performance. SSL is usually categorized into generative learning and contrastive learning [36]. Generative learning usually uses generative models to predict the labels of input data or generate new data samples. However, it is a pixel-level modeling method, making it challenging to establish the spatial structure relationships between modalities. By comparison, contrastive learning employs positive and negative samples to understand the consistency of images, enhancing its capability to differentiate between distinct images. Its superior efficacy and broad applicability have propelled it to the forefront of SSL [37].

By constructing potential space, contrastive learning makes similar samples close together and dissimilar samples far away. Numerous excellent algorithms for contrastive learning, such as SimCLR, bootstrap your own latent (BYOL) [38], momentum contrast (MoCo) [39], swap assignments between multiple views of the same image (SwAV) [40], and DINO, have been proposed in the CV field. Contrastive learning mainly includes the following four key modules.

1) *Data Augmentation:* This module is a crucial component in contrastive learning, targeting random enhancement of input data to generate positive and negative samples.
2) *Feature Extraction:* This module uses an encoder network $e(\cdot)$ to extract features from the expanded sample instances, which can be used to construct the feature representations of downstream tasks.
3) *Projection Head:* This module is a multilayer perceptron (MLP). It can prevent loss of information and enhance the quality of feature representations.
4) *Contrastive Loss:* This module uses a similarity metric to measure how close two input features are. Contrastive loss aims for similarity in positive pairs and dissimilarity in negative ones.

B. Multimodal Semantic Segmentation

With the rapid development of aerospace and astronautics, the RSISS plays an ever more significant role in many Earth observation tasks. Over the past decade, numerous researchers have developed many algorithms and models for RSISS using single-modal remote sensing data, yielding impressive results. However, training models by using single-modal remote sensing data, such as optical or SAR data, inevitably leads to the performance bottleneck in RSISS [41]. Therefore, some MRSISS methods have been proposed in the current remote sensing community.

Although the complexity of multimodal remote sensing data has posed great challenges in the development of more advanced multimodal algorithms and models, the remote sensing community has made some remarkable achievements [14]. Based on the fusion stage and motivation, we divide the multimodal semantic segmentation model into early, middle, and late fusion [42]. Early fusion's primary strategy involves merging raw multimodal data into various channels through concatenation [43]. We can use the individual semantic segmentation model to train the learning model end-to-end. Conversely, late fusion approaches concentrate on combining feature maps from various modalities at the decision-making layer. Specifically, they process each type of multimodal data in distinct streams. In the decoding phase, the features computed in these streams are brought into a unified space via fusion techniques (i.e., summing and weighted averaging), before passing through several convolutional layers [44]. The middle fusion methods are carefully crafted to leverage the advantages of both early and late fusion techniques. Generally, segmented networks access data through corresponding branches. Then, several extra modules are employed to calculate the modal weights and connect the encoder and decoder [42]. Hence, middle fusion networks are capable of adaptively creating a unified feature representation across different modalities, resulting in improved accuracy and robustness.

Recent research has explored the application of middle fusion model for multimodal semantic segmentation, using heterogeneous features of optical and SAR data. Hosseinpour et al. [45] proposed a deep cross-modal gated fusion network (CMGFNet) for the extraction of buildings from multimodal RSI. The CMGFNet extracts multilevel features from multimodal data through two separate encoders and effectively combines cross-modal feature fusion. Ren et al. [46] introduced DDHRNet, a dual-stream high-resolution network that seamlessly integrates optical and SAR data at the feature level within each branch, enabling effective utilization of complementary information from heterogeneous images. These multimodal networks have been used in RSISS, breaking the performance bottleneck and obtaining better semantic segmentation results than those using a single modality alone [47].

III. Methodology

A. Overall Network Architecture

Fig. 1 shows the multimodal contrastive learning paradigm. On this basis, we propose a novel multimodal contrastive learning framework based on tuple perturbation. In detail, this framework includes a pretrained stage TMCNet and a fine-tuning stage MSSNet, which are used to perform semantic segmentation on multimodal RSI.

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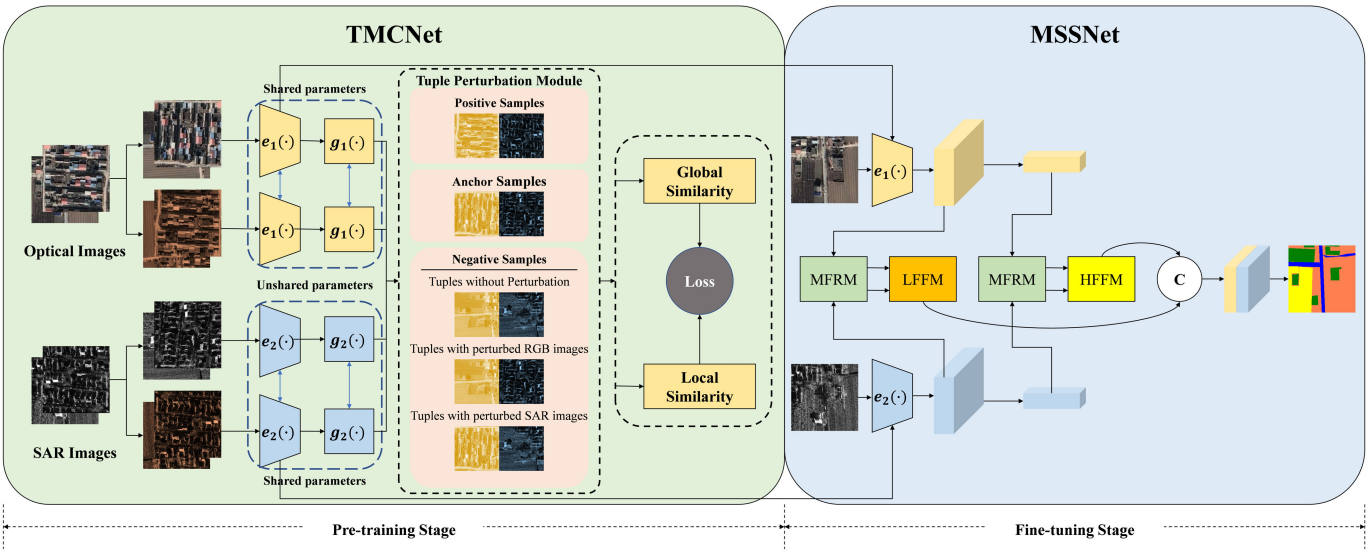


Fig. 2. Architecture of tuple perturbation-based contrastive learning framework for MRSISS.

During the pretraining stage, two parallel backbones without parameter sharing are employed to individually extract features from optical and SAR images. Generating complex negative samples is crucial for effective representation learning in contrastive learning, especially in the case of multimodal contrastive learning involving optical and SAR images. Consequently, we propose a tuple perturbation module, which can generate more challenging negative samples by disturbing optical or SAR features in the same scene, thus improving the ability of contrastive learning networks to distinguish samples.

In the fine-tuning stage, considering that the multimodal features have their own unique noisy information and that the feature of one modality has the potential to rectify and calibrate the noisy information of the other feature, we propose an MFRM to rectify the multimodal feature. After that, we design the MMFFM module that includes an HFFM of fusing the high-level features and an LFFM of fusing the low-level features. Ultimately, the decoder transforms feature maps of varying dimensions into the conclusive semantic map.

B. Tuple Perturbation-Based Multimodal Contrastive Learning Network

Given that contrastive learning improves by making positive samples similar and negative samples dissimilar, its core lies in the generation of positive and negative samples. On this basis, multimodal contrastive learning considers the differences and complementarities between different modalities and encourages the model to distinguish different samples from different modalities. We design the TMCNet to train the encoder of MSSNet, as shown in Fig. 2, which consists of the following five key parts.

1) Data Augmentation: To incentivize the model to acquire features, we generate four augmented views $\hat{x}$, $\tilde{x}$, $\hat{y}$, and $\tilde{y}$ from a given set of optical and SAR images x and y by data augmentation t_1 and t_2

$$\hat{x} = t_1(x), \quad \tilde{x} = t_2(x), \quad \hat{y} = t_1(y), \quad \tilde{y} = t_2(y). \tag{1}$$

In this study, t_1 denotes the process of random cropping and subsequently resizing to a constant resolution. Meanwhile, t_2 involves a sequence of enhancements, including the steps in t_1, along with random flipping, color distortion, noise introduction, and more.

2) Feature Extraction: The pseudo-Siamese encoder networks $e_1(\cdot)$ and $e_2(\cdot)$ are employed to extract global features from the optical and SAR augmented sample instances, respectively, enabling the construction of representations for downstream tasks

$$\begin{aligned} \hat{f}_{x_i} &= \mu\left(e_1\left(\hat{x}_i\right)\right), \quad \tilde{f}_{x_i} = \mu\left(e_1\left(\tilde{x}_i\right)\right) \\ \hat{f}_{y_i} &= \mu\left(e_2\left(\hat{y}_i\right)\right), \quad \tilde{f}_{y_i} = \mu\left(e_2\left(\tilde{y}_i\right)\right) \end{aligned} \tag{2}$$

where μ represents the global average pooling. In this work, $e_1(\cdot)$ and $e_2(\cdot)$ serve as the encoders of the DeepLabV3+.

3) Projection Head: A nonlinear projection function $g(\cdot)$ further processes the extracted features, mapping them to the layer z, at which the loss is computed

$$\begin{aligned} \hat{z}_{x_i} &= g\left(\hat{f}_{x_i}\right) = W^{(2)} R\left(W^{(1)} \hat{f}_{x_i}\right), \quad \tilde{z}_{x_i} = g\left(\tilde{f}_{x_i}\right) \\ \hat{z}_{y_i} &= g\left(\hat{f}_{y_i}\right) = W^{(2)} R\left(W^{(1)} \hat{f}_{y_i}\right), \quad \tilde{z}_{y_i} = g\left(\tilde{f}_{y_i}\right) \end{aligned} \tag{3}$$

where R is the rectified linear unit (ReLU).

4) Tuple Perturbation: Generating challenging negative samples is essential for effective representation learning in contrastive learning, especially in the case of multimodal contrastive learning for optical and SAR images, in which optical modality tends to dominate the learned representation. In order to alleviate the limitation of ignoring SAR modality and cooperation between modalities in contrastive learning, we propose a tuple perturbation module to generate negative samples, in which different modalities are different for different scenes. This prevents the network from becoming lazy and overly focusing on the optical modality.

Given a set of optical and SAR images, as illustrated in Fig. 3, we input the optical and SAR images into the pseudo-Siamese encoder to extract the features and then input the features into the tuple perturbation module.

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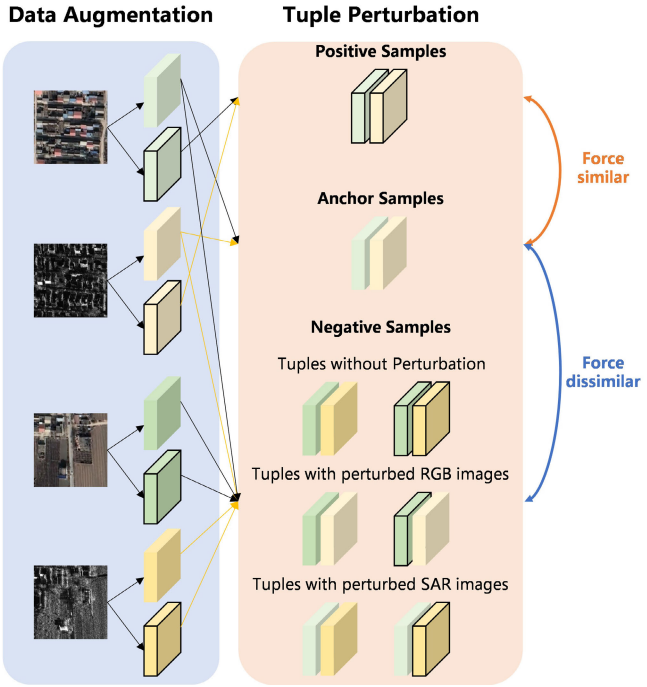


Fig. 3. Details of tuple perturbation module.

Specifically, the augmentation is applied to N optical and SAR sample pairs from a mini-batch, generating a total of $2N$ pairs. Each augmented pair originating from the same initial pair constitutes positive samples, while the remaining $2(N-1)$ pairs are treated as negatives. Building upon this foundation, the tuple perturbation module generates an additional $4(N-1)$ more challenging negative samples, of which the optical features in $2(N-1)$ sample pairs are consistent with those in the anchor sample, and their corresponding SAR features are perturbed; specifically speaking, SAR features are replaced by SAR features in different regions. Similarly, the other $2(N-1)$ sample pairs keep the SAR features unchanged and replace the optical features. Therefore, when the anchor sample is defined as $\hat{z}_{x_i} \otimes \hat{z}_{y_i}$, the positive sample is $\tilde{z}_{x_i} \otimes \tilde{z}_{y_i}$ and the negative samples are $\sum_{j=1,(j\neq i)}^{N}(\hat{z}_{x_j} \otimes \hat{z}_{y_j} + \tilde{z}_{x_j} \otimes \tilde{z}_{y_j}) + \sum_{j=1,(j\neq i)}^{N}(\hat{z}_{x_i} \otimes \hat{z}_{y_j} + \tilde{z}_{x_i} \otimes \tilde{z}_{y_j}) + \sum_{j=1,(j\neq i)}^{N}(\hat{z}_{x_j} \otimes \hat{z}_{y_i} + \tilde{z}_{x_j} \otimes \tilde{z}_{y_i})$, where $\otimes$ denotes the feature fusion.

5) Contrastive Loss: Contrastive loss encourages similarity between positive sample pairs and dissimilarity among negative sample pairs. From the initial N samples in a mini-batch, augmentation results in $4N-2$ samples. A positive pair arises from augmenting the same initial sample, while $4(N-1)$ samples compose the negative set. The global contrastive loss $\mathcal{L}_G$ is thus defined as

$$\mathcal{L}_G = \frac{1}{2N}\sum_{i=1}^{N}\left(\ell\left(\hat{m}_i + \tilde{m}_i\right) + \ell\left(\tilde{m}_i + \hat{m}_i\right)\right). \tag{4}$$

With

$$\ell\left(\hat{m}_i + \tilde{m}_i\right) = -\log \frac{\exp\left(\text{sim}\left(\hat{z}_{x_i} \otimes \hat{z}_{y_i}, \tilde{z}_{x_i} \otimes \tilde{z}_{y_i}\right)/\tau\right)}{\sum_{x\in\Lambda-}\exp\left(\text{sim}\left(\hat{z}_{x_i} \otimes \hat{z}_{y_i}, xy\right)/\tau\right)}$$

where xy represents the negative samples.

In the semantic segmentation dataset, a single image can contain a wide variety of land cover categories. Extracting only the global features of the entire image to measure and distinguish it can result in significant information loss. To address this, we have used a local contrastive learning module. Specifically, we select multiple nonoverlapping local regions, each with a size of $s_p \times s_p$, from N original images and introduce index labels to record the pixel positions, ensuring that the central positions of the two matched local regions correspond to each other in the original image. The local contrastive loss $\mathcal{L}_L$ is thus defined as

$$\mathcal{L}_L = \frac{1}{2N}\sum_{i=k}^{N_L}\left(\ell\left(\hat{m}_k + \tilde{m}_k\right) + \ell\left(\tilde{m}_k + \hat{m}_k\right)\right). \tag{5}$$

With

$$\ell\left(\hat{m}_i + \tilde{m}_i\right) = -\log \frac{\exp\left(\text{sim}\left(\hat{z}_{x_k} \otimes \hat{z}_{y_k}, \tilde{z}_{x_k} \otimes \tilde{z}_{y_k}\right)/\tau\right)}{\sum_{x\in\Lambda-}\exp\left(\text{sim}\left(\hat{z}_{x_k} \otimes \hat{z}_{y_k}, xy\right)/\tau\right)}.$$

In this work, the final loss is defined by combining the global and local contrastive losses

$$\mathcal{L} = \gamma \cdot \mathcal{L}_G + (1-\gamma)\cdot \mathcal{L}_L \tag{6}$$

where γ is set as a hyperparameter of 0.5, $\mathcal{L}_G$ represents the global contrastive loss, and $\mathcal{L}_L$ represents the local contrastive loss. Both $\mathcal{L}_G$ and $\mathcal{L}_L$ adopt the loss function $\mathcal{L}_C$.

C. Multimodal Semantic Segmentation Network

After the pretraining stage, the feature extractor produces better representation ability. Recent studies show that the attention mechanism is beneficial to the performance of the network [48], [49], [50]. To further improve the segmentation performance of the feature extractor, the MSSNet is proposed. The network takes independent optical and SAR images, along with their pixel-to-pixel corresponding labels, as the input. To extract more optical and SAR features from RSI, different levels of feature representations can be obtained using the different residual blocks of the pretrained feature extractor. In general, a feature extractor consists of some stages, each containing a sequence of residual blocks. In this scenario, low- and high-level feature representations can be extracted from different residual blocks of the pretrained feature extractor, allowing for the acquisition of feature information at different abstraction levels. Recognizing the features of different modalities have their own unique noise information, we design the MFRM to rectify the optical and SAR features to minimize the impact of noise on segmentation results. In the feature fusion stage, we separately design the LFFM and HFFM for the feature fusion of optical and SAR images after feature rectification, so as to enhance information interaction and improve segmentation performance. Finally, the deconvolution operation is used to sample the fused features to the original input size and merge them to produce the segmentation map. Details of the MFRM, LFFM, and HFFM will be shown in the following.

1) Multimodal Feature Rectification Module: It has been found that features originating from different modalities have their own unique noise information, and the features of one modality can be used to rectify and calibrate the noise information of another modality's features [1]. Accordingly,

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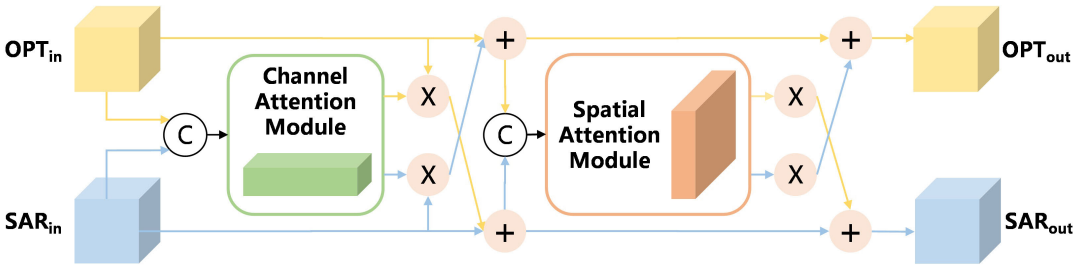


Fig. 4. Details of MFRM.

we use the MFRM to rectify the information extracted in the feature extraction process. To deal with all kinds of noises and uncertainties in different modalities, MFRM processes features in two dimensions—channel and space—to enhance multimodal feature extraction and interaction.

As shown in Fig. 4, first, we input the features of two modalities, which are $\mathrm{OPT_{in}}(\mathrm{OPT_{in}} = \mathrm{OPT_{in}^{low}}$ or $\mathrm{OPT_{in}^{high}})$ and $\mathrm{SAR_{in}}(\mathrm{SAR_{in}} = \mathrm{SAR_{in}^{low}}$ or $\mathrm{SAR_{in}^{high}})$, into the channel attention module to get two attention weight maps W_{OPT}^{C} and W_{SAR}^{C}. In the channel attention module, max pooling and average pooling are applied to $\mathrm{OPT_{in}}$ and $\mathrm{SAR_{in}}$ along the channel dimension to obtain four resulting vectors. Next, we concatenate them to get feature Y and apply an MLP, which is followed by a sigmoid function, to get the weight W^{C} from Y, which consists of W_{OPT}^{C} and W_{SAR}^{C}

$$\begin{aligned} Y = \big[&\mathrm{Avg}(\mathrm{OPT_{in}}), \mathrm{Max}(\mathrm{OPT_{in}}) \\ &\mathrm{Avg}(\mathrm{SAR_{in}}), \mathrm{Max}(\mathrm{SAR_{in}})\big] \end{aligned} \tag{7}$$

$$W_{\mathrm{OPT}}^{C}, W_{\mathrm{SAR}}^{C} = F_{\mathrm{split}}(\sigma(F_{\mathrm{MLP}}(Y))) \tag{8}$$

where input concatenation is denoted by $[,]$ in the channel dimension and σ represents the sigmoid function. The channel-wise feature rectification is formulated as follows:

$$\begin{aligned} \mathrm{OPT_{out}^{1}} &= \mathrm{OPTin} + \lambda_C W_{\mathrm{SAR}}^{C} \otimes \mathrm{SARin} \\ \mathrm{SAR_{out}^{1}} &= \mathrm{SAR_{in}} + \lambda_C W_{\mathrm{OPT}}^{C} \otimes \mathrm{OPT_{in}} \end{aligned} \tag{9}$$

where $\otimes$ denotes the channel-wise multiplication and λ_C is a hyperparameter and we set it as 0.5.

Because the channel attention module focuses on learning global weights for rectification, we further introduce a spatial attention module to learn local weights for rectification. The two modalities' features $\mathrm{OPT_{out}^{1}}$ and $\mathrm{SAR_{out}^{1}}$ will be concatenated and input into the spatial attention module to get two attention weight maps: W_{OPT}^{S} and W_{SAR}^{S}. The spatial attention module contains two convolution layers and a ReLU function. We input optical and SAR features into this module to get a weight map, which consists of W_{OPT}^{S} and W_{SAR}^{S}

$$F = \mathrm{Conv_{1\times1}}(\mathrm{ReLU}(\mathrm{Conv_{1\times1}}[\mathrm{OPT_{in}}, \mathrm{SAR_{in}}])) \tag{10}$$

$$W_{\mathrm{OPT}}^{S}, W_{\mathrm{SAR}}^{S} = F_{\mathrm{split}}(\sigma(F)). \tag{11}$$

Just like channel-wise rectification, spatial-wise rectification is expressed as

$$\begin{aligned} \mathrm{OPT_{out}} &= \mathrm{OPT_{out}^{1}} + \lambda_S W_{\mathrm{SAR}}^{S} \oplus \mathrm{SAR_{in}} \\ \mathrm{SAR_{out}} &= \mathrm{SAR_{out}^{1}} + \lambda_S W_{\mathrm{OPT}}^{S} \oplus \mathrm{OPT_{in}} \end{aligned} \tag{12}$$

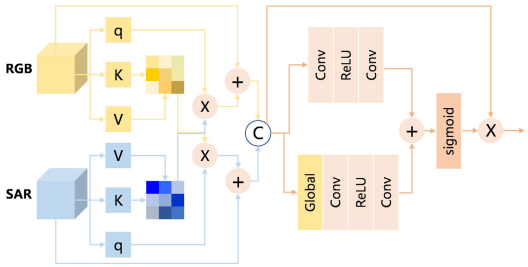


Fig. 5. Details of LFFM.

where $\oplus$ denotes spatial-wise multiplication and λ_S is a hyperparameter and we set it as 0.5.

$\mathrm{OPT_{out}}$ $(\mathrm{OPT_{out}} = \mathrm{OPT_{out}^{low}}$ or $\mathrm{OPT_{out}^{high}})$ and $\mathrm{SAR_{out}}$ $(\mathrm{SAR_{out}} = \mathrm{SAR_{out}^{low}}$ or $\mathrm{SAR_{out}^{high}})$, having undergone comprehensive rectification, will proceed to the next stage for feature fusion.

2) Low-Level Feature Fusion Module: LFFM fuses low-level optical and SAR features, which are rectified by the MFRM, aiming at more compactly integrating multimodal information and constructing global correlation of multimodal images. Fig. 5 shows the details of LFFM. Given the outputs $\mathrm{OPT_{out}^{low}}$ and $\mathrm{SAR_{out}^{low}}$ of MFRM, we input them into LFFM to get the fusion feature F_{LFFM}. First, the interactive attention maps $f_{\mathrm{attention}}^{\mathrm{OPT-low}}$ and $f_{\mathrm{attention}}^{\mathrm{SAR-low}}$ are generated from $\mathrm{OPT_{out}^{low}}$ and $\mathrm{SAR_{out}^{low}}$, respectively. A convolution is utilized to convert the optical features into three identical feature maps: Q_{OPT}, K_{OPT}, and V_{OPT} with an equal number of channels. Similarly, we obtain three identical feature maps: Q_{SAR}, K_{SAR}, and V_{SAR} from the SAR features. The formulas are as follows:

$$\begin{aligned} Q_{\mathrm{OPT}} &= \mathrm{Conv_{1\times1}}\left(\mathrm{OPT_{out}^{low}}\right) \\ K_{\mathrm{OPT}} &= \mathrm{Conv_{1\times1}}\left(\mathrm{OPT_{out}^{low}}\right) \\ V_{\mathrm{OPT}} &= \mathrm{Conv_{1\times1}}\left(\mathrm{OPT_{out}^{low}}\right) \end{aligned} \tag{13}$$

$$\begin{aligned} Q_{\mathrm{SAR}} &= \mathrm{Conv_{1\times1}}\left(\mathrm{SAR_{out}^{low}}\right) \\ K_{\mathrm{SAR}} &= \mathrm{Conv_{1\times1}}\left(\mathrm{SAR_{out}^{low}}\right) \\ V_{\mathrm{SAR}} &= \mathrm{Conv_{1\times1}}\left(\mathrm{SAR_{out}^{low}}\right). \end{aligned} \tag{14}$$

To derive the self-attention of each data source, the transposes of V and K are multiplied and subsequently passed through the SoftMax function. The formulas are as follows:

$$\begin{aligned} f_{\mathrm{attention}}^{\mathrm{OPT-low}} &= \mathrm{softmax}\left(V_{\mathrm{OPT}}{}^{T} K_{\mathrm{OPT}}\right) \\ f_{\mathrm{attention}}^{\mathrm{SAR-low}} &= \mathrm{softmax}\left(V_{\mathrm{SAR}}{}^{T} K_{\mathrm{SAR}}\right). \end{aligned} \tag{15}$$

In addition, the output is the result of the cross-attention process where interactive attention maps are multiplied with

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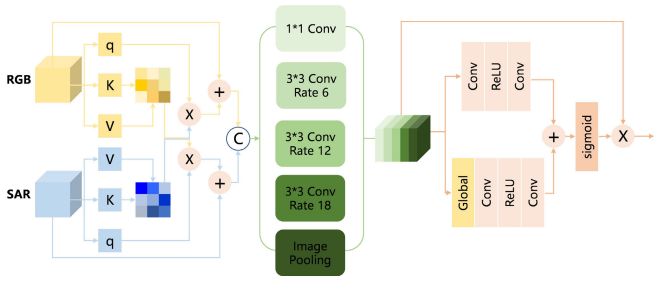


Fig. 6. Details of HFFM.

the feature map from a different modality pathway, and it is given by

$$f_{\mathrm{LFFM}}^{\mathrm{OPT-low}} = \left(Q_{\mathrm{OPT}} \otimes f_{\mathrm{attention}}^{\mathrm{SAR-low}}\right) \oplus \mathrm{OPT}_{\mathrm{out}}^{\mathrm{low}}$$
$$f_{\mathrm{LFFM}}^{\mathrm{SAR-low}} = \left(Q_{\mathrm{SAR}} \otimes f_{\mathrm{attention}}^{\mathrm{OPT-low}}\right) \oplus \mathrm{SAR}_{\mathrm{out}}^{\mathrm{low}}. \quad (16)$$

Note that $f_{\mathrm{LFFM}}^{\mathrm{OPT-low}}$ and $f_{\mathrm{LFFM}}^{\mathrm{SAR-low}}$ denote the feature maps with cross-attention augmentation. Moreover, the concatenated result of $f_{\mathrm{LFFM}}^{\mathrm{OPT-low}}$ and $f_{\mathrm{LFFM}}^{\mathrm{SAR-low}}$ enhances the representation of fused features, as illustrated by

$$f_{\mathrm{LFFM}} = \left[f_{\mathrm{LFFM}}^{\mathrm{OPT-low}}, f_{\mathrm{LFFM}}^{\mathrm{SAR-low}}\right] \quad (17)$$

where [,] denotes the concatenating of the input in the channel dimension. Then, f_{LFFM} is provided to the multiscale channel attention module [51]. It can be formulated as follows:

$$F_{\mathrm{LFFM}} = f_{\mathrm{LFFM}} \otimes \left(\mathrm{sigmod}\left(f_{\mathrm{global}} \oplus f_{\mathrm{local}}\right)\right). \quad (18)$$

With

$$f_{\mathrm{global}} = \mathrm{Conv}_{1\times1}(\mathrm{ReLU}(\mathrm{Conv}_{1\times1}(G(f_{\mathrm{LFFM}}))))$$
$$f_{\mathrm{local}} = \mathrm{Conv}_{1\times1}(\mathrm{ReLU}(\mathrm{Conv}_{1\times1}(f_{\mathrm{LFFM}}))).$$

3) High-Level Feature Fusion Module: HFFM fuses high-level optical and SAR features, which are rectified by the MFRM, aiming at more compactly integrating multimodal information and constructing global correlation of multimodal images. Fig. 6 shows the details of HFFM. Given the outputs $\mathrm{OPT}_{\mathrm{out}}^{\mathrm{high}}$ and $\mathrm{SAR}_{\mathrm{out}}^{\mathrm{high}}$ of MFRM, we input them into HFFM to get the fusion feature F_{HLFFM}. First, a more powerful representation of fused feature f_{HFFM} is generated from $\mathrm{OPT}_{\mathrm{out}}^{\mathrm{high}}$ and $\mathrm{SAR}_{\mathrm{out}}^{\mathrm{high}}$. Similar to f_{LFFM}, f_{HFFM} is defined as follows:

$$f_{\mathrm{HFFM}} = \left[f_{\mathrm{HFFM}}^{\mathrm{OPT-high}}, f_{\mathrm{HFFM}}^{\mathrm{SAR-high}}\right]. \quad (19)$$

Second, in contrast to LFFM, we introduce the ASPP module into HFFM. Considering the lower resolution of high-level features obtained through multiple downsampling, which cannot effectively deal with objects of different scales, we introduce the ASPP module before feature enhancement to get $f_{\mathrm{HFFM}}^{\mathrm{ASPP}}$, which is defined as follows:

$$f_{\mathrm{HFFM}}^{\mathrm{ASPP}} = \mathrm{ASPP}(f_{\mathrm{HFFM}}). \quad (20)$$

The ASPP module is responsible for performing multiple parallel dilated convolution operations, each with a different sampling rate. This allows for the extraction of richer feature representations in various receptive fields and enhances the ability to capture semantic information across various scales.

Then, we input $f_{\mathrm{HFFM}}^{\mathrm{ASPP}}$ into the multiscale channel attention module to get the fusion feature F_{HFFM}. It is defined as follows:

$$F_{\mathrm{HFFM}} = f_{\mathrm{HFFM}} \otimes \left(\mathrm{sigmod}\left(f_{\mathrm{global}} \oplus f_{\mathrm{local}}\right)\right). \quad (21)$$

With

$$f_{\mathrm{global}} = \mathrm{Conv}_{1\times1}(\mathrm{ReLU}(\mathrm{Conv}_{1\times1}(G(f_{\mathrm{HFFM}}))))$$
$$f_{\mathrm{local}} = \mathrm{Conv}_{1\times1}(\mathrm{ReLU}(\mathrm{Conv}_{1\times1}(f_{\mathrm{HFFM}}))).$$

Low- and high-level feature maps are created through the feature fusion module, which directs MSSNet to concentrate on the region of interest and enhances its ability to fuse features between multimodal data. Ultimately, two feature maps are input into the decoder to produce the prediction map.

IV. Experiments and Results

A. Data Description

We assess the proposed TMCNet alongside other self-supervised approaches, as well as the proposed MSSNet alongside other multimodal semantic segmentation approaches on two multimodal datasets for MRSISS.

1) YESeg-OPT-SAR Dataset: This dataset is a sufficiently labeled RSI for land use segmentation, which consists of 2231 high-resolution optical images and SAR images. Both the spatial size of optical and SAR images is 256 × 256 at a 0.5-m resolution. The images in the dataset are completely labeled with seven pixel-level categories, including background, bare land, vegetation, trees, houses, water, roads, and others. The YESeg-OPT-SAR dataset is presented on https://github.com/yeyuanxin110/YESeg-OPT-SAR.

2) WHU-OPT-SAR Dataset: This dataset is a large-scale optical and SAR joint land use segmentation dataset, which is built by using the optical images of GaoFen-1 satellite and the SAR image of GaoFen-3 satellite in the same area [46]. The dataset consists of 100 images, spanning an area totaling 51448.56 km^2, at a 5-m resolution. Enhanced with pixel-level annotations, these images are ideal for deep learning-driven land use segmentation tasks. Seven labels categorize the pixels into distinct classes such as farmland, urban, rural, aquatic, forested areas, roads, and miscellaneous features. The WHU-OPT-SAR dataset, alongside its corresponding annotations, is accessible at https://github.com/AmberHen/WHU-OPT-SAR-dataset.

B. Evaluation Metrics and Implementation Details

1) Evaluation Metrics: To evaluate the proposed framework, we apply overall accuracy (OA) and Kappa coefficients to determine the accuracy across the test datasets for downstream tasks. These metrics are described as

$$\mathrm{OA} = \frac{\mathrm{TP}}{N} \quad (22)$$

$$\mathrm{Kappa} = \frac{\mathrm{OA} - p_e}{1 - p_e} \quad (23)$$

where TP represents the total number of correctly predicted pixels, which is the total number of pixels. The pixel-wise accuracy p_e is calculated as $(a_1 \times b_1 + \cdots + a_c \times b_c / N \times N)$, where a_c is the actual number of pixels of class c and b_c represents the number of predicted pixels of class c.

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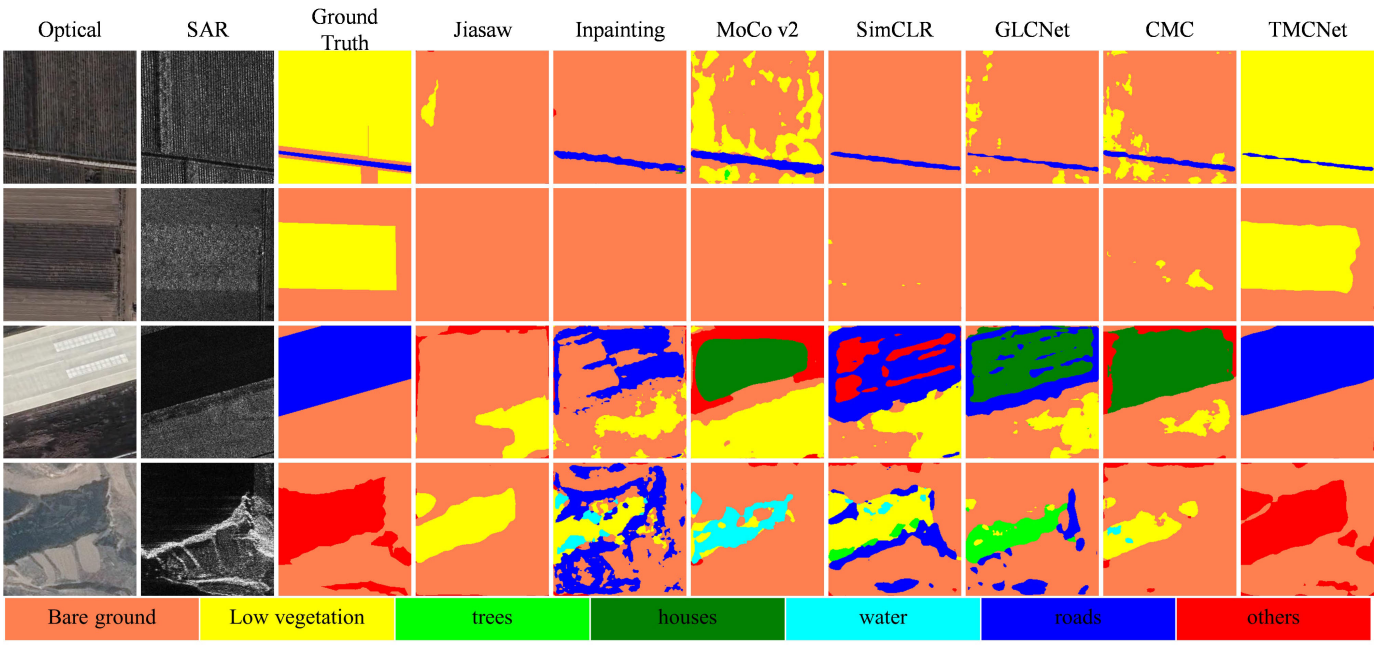


Fig. 7. Examples of visualization results on the YESeg-OPT-SAR dataset on TMCNet and other self-supervised networks.

TABLE I
SAMPLE SIZE OF TWO TYPICAL MULTIMODAL DATASETS (256 × 256 PIXELS)

Dataset	Pre-training	Train	Test
YESeg-OPT-SAR	2231	800	1431
WHU-OPT-SAR	3528	800	2728

2) Implementation Details: All experiments were conducted in Pytorch 1.11.0 basis on python 3.7 on an NVIDIA GeForce RTX 3090 GPU and an Intel Core i7-12700KF CPU. To assess the performance of our proposed multimodal contrastive learning framework based on tuple perturbation, we compare TMCNet with other typical contrastive learning models (i.e., Jigsaw, Inpainting, MoCo v2, SimCLR, GLCNet [52], and contrastive multiview coding (CMC) [30]) in the self-supervised pretraining stage and compare MSSNet with other classic MSSNets (i.e., MCANet, DDHRNet, and CMGFNet) in the fine-tuning stage. For the two datasets, 800 pairs of data are randomly chosen for training, while the remaining data serve as test samples in our experiments. Table I displays the specifics regarding the training and test samples.

a) Implementation details on TMCNet: In order to control the variables and ensure the reliability of the experiment, we apply the same MSSNet in the downstream task, and its feature extractor is ResNet50. Therefore, Jigsaw, Inpainting, SimCLR, MoCo v2, GLCNet, CMC, and TMCNet are only designed to train the ResNet50 feature extractor in the contrastive learning pretraining stage.

We use the Adam optimizer to optimize the proposed model. During the pretraining stage, the batch size is set to 32, the initial learning rate is set to 0.01 with the cosine decay schedule, and 100 epochs are executed. The pretraining model takes nearly 120 h. Moreover, for the proposed TMCNet, we fuse optical features and SAR features in an appropriate ratio. The model achieving the minimum loss during the pretraining stage is preserved for subsequent tasks.

b) Implementation details on MSSNet: In the pretraining stage, we use TMCNet to pretrain the feature extractor, while in the downstream multimodal semantic segmentation task, we use a variety of MSSNets for comparative experiments. For multimodal semantic segmentation training, a sparse dataset comprising 800 pairs of self-supervised data is employed. The training process with this dataset takes approximately 5 h. The Adam optimizer is utilized for refining the suggested model. In the fine-tuning phase, the batch size is set to 32, while the initial learning rate is set to 0.01 with the cosine decay schedule, and 100 epochs are executed.

C. Accuracy Analysis on TMCNet

This section employs the test datasets from Section IV-A to contrast TMCNet with alternative approaches. We perform both qualitative and quantitative evaluations on the two datasets for the mentioned methods.

1) Qualitative Accuracy Analysis: In this section, we qualitatively evaluate the performance of the proposed TMCNet on MRSISS tasks on multiple datasets with limited labels and compare it with other contrastive learning methods. Figs. 7 and 8, respectively, show the examples of visualization results of self-supervised tasks on both kinds of datasets. The other self-supervised semantic segmentation methods produce rich segmentation maps. However, it is challenging to balance the optical and SAR features where the optical features are interfered by SAR features, resulting in a decrease in the

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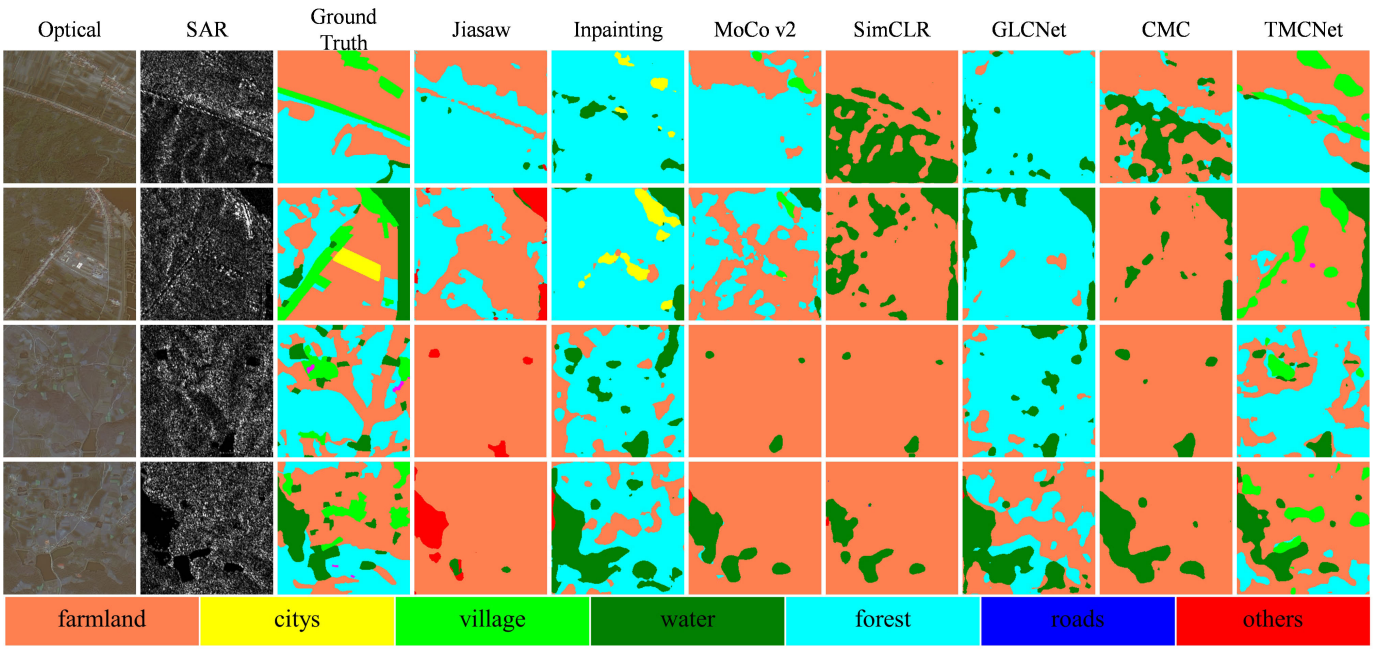


Fig. 8. Examples of visualization results on the WHU-OPT-SAR dataset on TMCNet and other self-supervised networks.

TABLE II
OA AND KAPPA VALUES ON THE YESEG-OPT-SAR DATASET (IN %) ON OTHER SELF-SUPERVISED NETWORKS

Method	OA	Kappa	User's Accuracy						
			Bare ground	Low vegetation	trees	houses	water	roads	others
Jiasaw	63.58	27.67	66.65	55.37	57.23	48.14	63.77	21.81	22.24
Inpainting	52.3	22.45	66.46	41.3	**58.91**	49.89	37.49	14.8	11.27
MoCo v2	56.33	20.49	65.08	44.63	55.58	59.49	27.03	39.24	7.43
SimCLR	59.93	37.38	75.31	64.43	52.43	61.2	72.24	22.62	6.94
GLCNet	62.58	32.63	68.13	72.39	57.16	63.35	87.27	19.06	34.67
CMC	57.7	26.93	67.69	62.94	49.75	60.89	44.57	47.89	6.47
TMCNet(Proposed)	**79.82**	**65.73**	**84.63**	**76.44**	57.91	**73.77**	**96.87**	**75.55**	**46.9**

accuracy of segmentation maps. Although CMC considers the problem of multimodal features, it ignores that it is difficult for a single feature encoder to adapt to multimodal data at the same time, resulting in the accuracy improvement not obvious. Fully considering the differences between optical and SAR images, our TMCNet uses the pseudo-Siamese network, which uses unshared parameters, to extract the optical features and SAR features, and uses tuple perturbation module to improve the performance of the feature extraction network, thus fully utilizing the information in optical and SAR images to improve the segmentation accuracy. Moreover, compared with other methods, the segmentation map generated by TMCNet can provide more realistic land cover labels and superior spatial aggregated results. The ground objects fail to be well identified by other methods due to the SAR images seriously affected by noise. However, the proposed TMCNet is capable of eliminating the influence of noise covering from the visual perspective. In general, our proposed model demonstrates improved capability in distinguishing land cover categories and producing more realistic segmentation maps.

2) Quantitative Accuracy Analysis: Tables II and III report the quantitative OA, kappa, and the accuracy of each class using the proposed method as well as the other self-supervised methods for the optical and SAR data, respectively, for the YESeg-OPT-SAR dataset and WHU-OPT-SAR dataset. The YESeg-OPT-SAR dataset has a high resolution of 0.5 m, while the WHU-OPT-SAR dataset has a low resolution of 5 m. The evaluation data show that the proposed TMCNet outperforms the rest by obtaining the highest OA and kappa and outperforms others in most class-wise segmentation accuracies. Each method will be further discussed next.

a) Jigsaw: The employed self-supervised technique forms its task through puzzle-solving exercises. An image is segmented into several pieces that are then randomized. Afterward, they are fed into a CNN to discern the contextual links between these segments. As can be seen from Tables II and III, compared with the other contrastive learning methods, Jigsaw has a good overall effect in processing RSIs, its OA is high, and Kappa is average. From Figs. 7 and 8, we can see that its ability to distinguish complex ground objects has no obvious advantage over other methods.

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TABLE III
OA AND KAPPA VALUES ON THE WHU-OPT-SAR DATASET (IN %) ON TMCNET AND OTHER SELF-SUPERVISED NETWORKS

Method	OA	Kappa	User's Accuracy						
			farmland	city	village	water	forest	road	others
Jiasaw	74.49	58.94	62.48	**91.59**	57.58	72.18	88.09	0	0
Inpainting	74.95	59.63	76.96	60.96	**92.03**	69.37	76.4	7.76	14.21
MoCo v2	77.47	64	72.95	61.89	85.92	73.92	82.75	9.63	9.82
SimCLR	67.41	50.97	61.61	88.85	48.31	27.99	90.01	6.46	3.74
GLCNet	73.28	56.32	73.72	74.72	92	59.49	75.06	9.36	2.77
CMC	76.46	63.8	68.99	78.16	76.3	50.5	87.08	2.11	0
TMCNet(Proposed)	**82.79**	**73.37**	**78.31**	72.4	66.04	**74.99**	**90.03**	**60.21**	**24.83**

b) Inpainting: Inpainting stands as a common technique for generating self-supervised signals, centering around the concept of image reconstruction. Initially, a certain segment of the image is removed. Subsequently, the CNN model undergoes training to reconstruct the original image from its impaired state. This enables the CNN model can learn the context information. As can be seen from Tables II and III, compared with the other contrastive learning methods, the accuracy of inpainting on the YESeg-OPT-SAR dataset is poor, and it is impossible to identify the ground objects with inconspicuous features. In contrast, the accuracy of inpainting on the WHU-OPT-SAR dataset is similar to the other contrastive learning methods. It can also be seen from Figs. 7 and 8 that it is easy to mistake the ground objects and cannot accurately identify the water body because it is easily affected by SAR image noise.

c) MoCo v2: MoCo v2 builds upon instance-based contrastive learning, striving for a broader scope of negative samples that surpass typical batch sizes. To achieve this, it introduces a dynamic queue for storing negative sample features and employs a momentum update mechanism for the encoder, mitigating representation consistency issues caused by rapid encoder adjustments. From Tables II and III, we can see that MoCo v2 cannot make full use of optical and SAR features, and its performance on both datasets is greatly affected by the quality of image labels. From Figs. 7 and 8, we can see that there are a lot of errors in the segmentation map.

d) SimCLR: The SimCLR method also employs instance-wise contrastive learning, which promotes the similarity of positive samples enhanced from a single sample and the dissimilarity of negative samples enhanced from various samples within a mini-batch. It can be seen from Tables II and III that compared with MoCo v2, the performance of SimCLR is improved on some datasets, but the optical and SAR features are not balanced, and its segmentation accuracy is poor. From Figs. 7 and 8, we can see that it is difficult to identify complex ground objects due to the unfavorable effects of SAR features.

e) GLCNet: The GLCNet method, like SimCLR, is based on the idea of instance-wise contrastive learning. Compared with SimCLR, it emphasizes the equilibrium between global and local feature learning in RSISS tasks for RSIs. As can be seen from Tables II and III, the performance of GLCNet has some limitations on the WHU-OPT-SAR dataset, but it has been improved on the YESeg-OPT-SAR dataset. From Figs. 7 and 8, it can be seen that the performance of GLCNet in road identification is significantly reduced, and there are obvious errors in the segmentation of other categories.

f) CMC: It is a deep learning method, and its goal is to train the shared information characteristics of multiple sensing perspectives by maximizing the mutual information of different perspectives in the same scene. Replacing different perspectives with different modalities can be used in MRSISS tasks, and the more modalities, the better the effect. As can be seen from Tables II and III, the performance of CMC in processing multimodal datasets has been significantly improved, especially on the WHU-OPT-SAR dataset. From Figs. 7 and 8, we can see that the segmentation of ground objects can be roughly judged, but the integrity of ground objects cannot be guaranteed.

g) TMCNet: The information provided in Tables II and III illustrates that TMCNet has achieved the best accuracy, and its performance has been significantly improved on the YESeg-OPT-SAR dataset and the WHU-OPT-SAR dataset. For example, on the WHU-OPT-SAR dataset, its comprehensive accuracy far exceeds all existing contrastive learning methods, although it is still difficult to identify village and other categories. For the YESeg-OPT-SAR dataset, TMCNet has more obvious advantages and achieves the best results in all categories. It can be clearly observed from Fig. 7 that TMCNet achieves the best performance, which can accurately identify low vegetation and other ground objects. From Fig. 8, our proposed model performs poorly in distinguishing between "city" and "village," particularly when the features of these areas are very similar to those of the surrounding "bare ground." Although TMCNet misclassifies ground objects, it is still superior to other methods. The above experimental results prove that TMCNet can be applied to various types of MRSISS tasks and adapted to datasets with different resolutions. In comparison to other self-supervised methods, TMCNet obtains the most comprehensive and accurate segmentation results overall.

D. Accuracy Analysis on MSSNet

In this section, we utilize the test datasets from Section IV-A to compare MSSNet with other methods. Both qualitative

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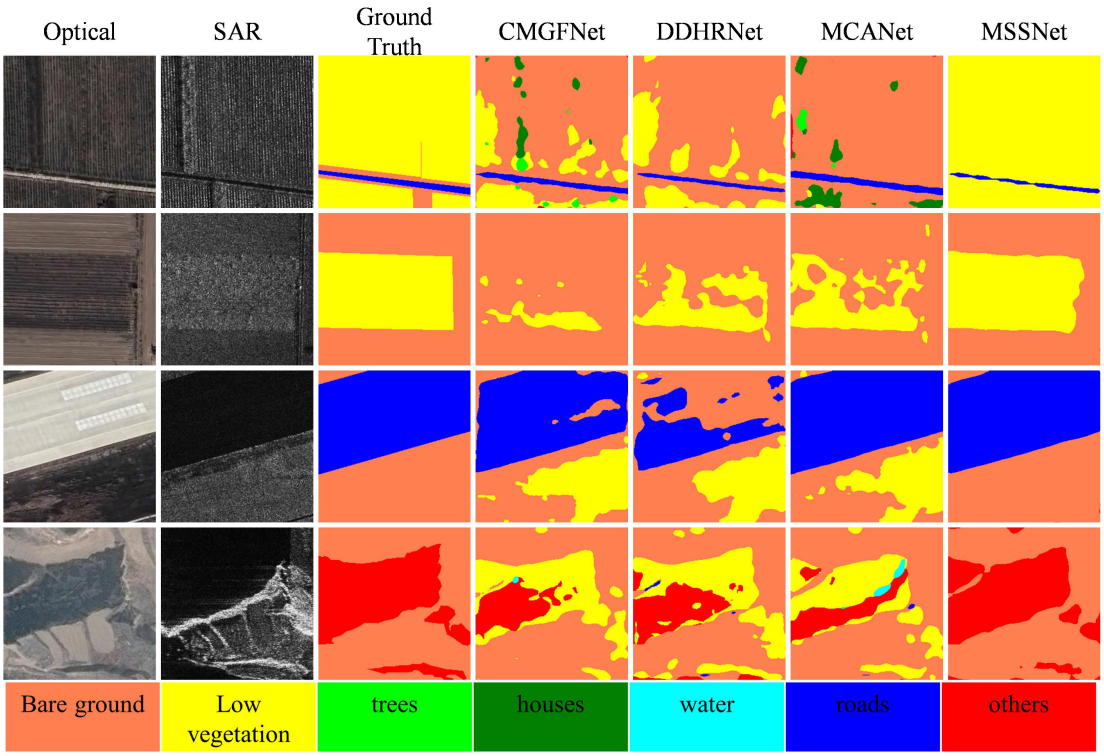


Fig. 9. Examples of visualization results on the YESeg-OPT-SAR dataset on MSSNet and other MSSNets.

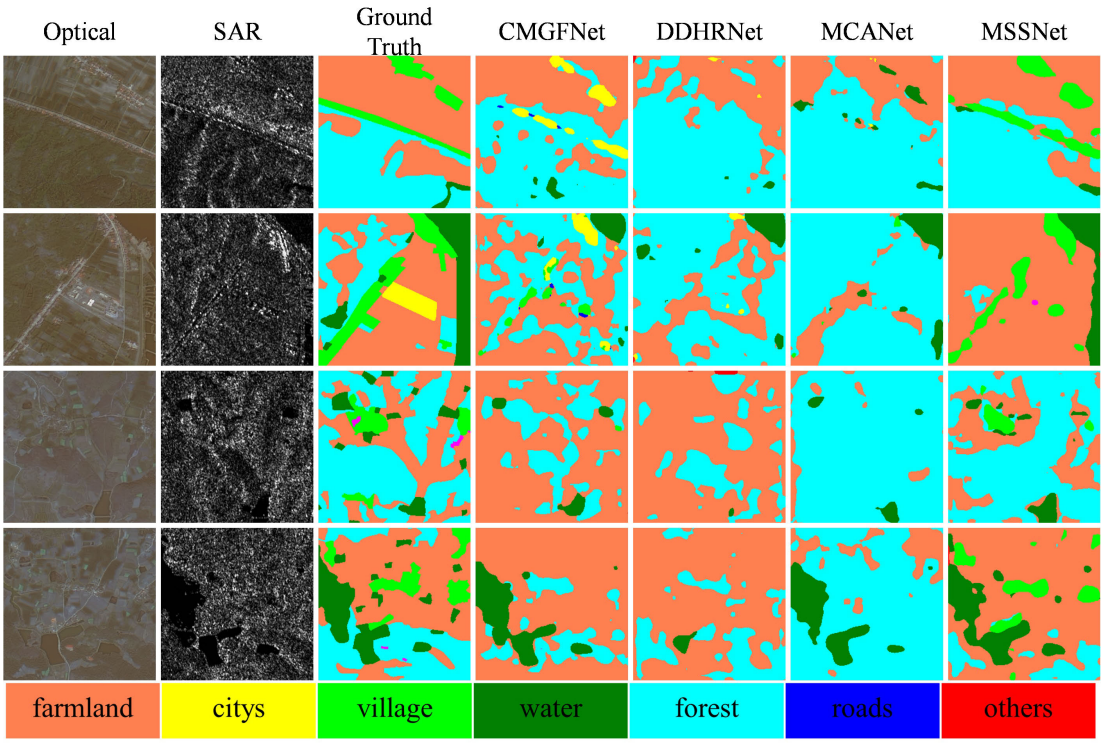


Fig. 10. Examples of visualization results on the WHU-OPT-SAR dataset on MSSNet and other MSSNets.

and quantitative analyses are conducted separately for these methods on the two datasets.

1) Qualitative Accuracy Analysis: We qualitatively evaluate the performance of the proposed MSSNet on MRSISS tasks on multiple datasets with limited labels, comparing it to other MSSNets. Figs. 9 and 10, respectively, show examples of visualized results of multimodal semantic segmentation tasks on the YESeg-OPT-SAR dataset and the WHU-OPT-SAR dataset. Although other multimodal semantic segmentation methods can generate a significant number of meaningful segmentation maps, the segmentation accuracy is not high enough because of the failure to deeply explore and integrate the

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TABLE IV
OA AND KAPPA VALUES ON THE YESEG-OPT-SAR DATASET (IN %) ON MSSNET AND OTHER MSSNETS

Method	OA	Kappa	User's Accuracy						
			Bare ground	Low vegetation	trees	houses	water	roads	others
CMGFNet	78.72	62.88	81.59	76.32	**68.86**	71.97	84.63	75.02	46.62
DDHRNet	77.84	61.01	80.77	73.92	66.38	72.67	77.17	63.43	59.4
MCANet	77.18	60.87	82.99	69.03	65.62	64.44	80.53	65.72	**61.86**
MSSNet(Proposed)	**79.82**	**65.73**	**84.63**	**76.44**	57.91	**73.77**	**96.87**	**75.55**	46.9

TABLE V
OA AND KAPPA VALUES ON THE WHU-OPT-SAR DATASET (IN %) ON MSSNET AND OTHER MSSNETS

Method	OA	Kappa	User's Accuracy						
			farmland	city	village	water	forest	road	others
CMGFNet	78.15	65.27	77.28	58.65	81.84	70.39	81.42	18.29	13.2
DDHRNet	74.76	59.94	70.59	53.11	87.16	52.52	82.63	16.82	0
MCANet	75.46	60.37	77.06	58.75	**91.7**	69.28	76.91	15.13	0
MSSNet(Proposed)	**82.79**	**73.37**	**78.31**	**72.4**	66.04	**74.99**	**90.03**	**60.21**	**24.83**

features of optical and SAR images. Taking full account of the shared and different features of optical and SAR images, based on multimodal feature correction, MSSNet fuses low- and high-level features, respectively, to improve the segmentation accuracy.

2) Quantitative Accuracy Analysis: Tables IV and V report the quantitative OA, kappa, and the accuracy of each class using the proposed model as well as the other multimodal semantic segmentation methods for the optical and SAR data, respectively, for the YESeg-OPT-SAR dataset and WHU-OPT-SAR dataset. The evaluation data show that the proposed MSSNet not only achieves the highest OA and kappa but also outperforms other multimodal semantic segmentation methods in most class-wise segmentation accuracies. Each method will be further discussed next.

a) CMGFNet: It employs dual encoders to capture multitiered features from diverse modalities, providing a pair of fusion techniques to amalgamate traits from two sources. It introduces the gated fusion module (GFM) to effectively merge cross-modal features. Moreover, it adopts a top–down approach to blend deep, high-level attributes with more superficial low-level features for multitiered fusion. Tables IV and V show that the performance of CMGFNet is acceptable. From Figs. 9 and 10, we can see that it has strong limitations.

b) DDHRNet: It intricately merges features from SAR and optical sources within each of its branches, leveraging the disparate data to enhance the segmentation quality, unaffected by cloud coverage. It incorporates a multimodal squeeze-and-excitation (MSE) mechanism to amalgamate these characteristics. This module outperforms standard techniques, offering a performance boost ranging from 1% to 5%. Tables IV and V demonstrate the limitations of DDHRNet in multimodal semantic segmentation. Figs. 9 and 10 reveal that DDHRNet does not improve the shortcomings of CMGFNet.

c) MCANet: It, inspired by the renowned Deeplabv3+ framework, is designed to classify land use utilizing both optical and SAR imagery. This network processes these images and their pixel-level annotations independently. From Tables IV and V, it can be seen that the performance of MCANet has not been improved compared with DDHRNet and CMGFNet. From Figs. 9 and 10, we can see that it also fails to improve the shortcomings of other schemes.

d) MSSNet: As can be seen from Tables IV and V, MSSNet obtains the highest accuracy on the YESeg-OPT-SAR dataset and the WHU-OPT-SAR dataset. From Fig. 9, we can see that the segmentation visualized result is excellent; in particular, the low vegetation will not be wrongly classified. From Fig. 10, we can see that it can accurately identify the ground object on the WHU-OPT-SAR dataset, but it is difficult for other MSSNets to identify the ground object.

The above experimental results demonstrate that MSSNet can be applied to various types of MRSISS tasks and can adapt to datasets with different resolutions. Comparing other multimodal semantic segmentation methods, MSSNet provides the most comprehensive and accurate segmentation results overall.

E. Ablation Study

1) Effectiveness of Tuple Perturbation Module: In Section III, we have introduced the specific architecture of tuple perturbation module designed for TMCNet. This module carries out tuple perturbation, combining optical modal perturbation and SAR modal perturbation. To evaluate the effects of various modal perturbation on TMCNet methods, we conduct experiments with no perturbation, optical modal perturbation, SAR modal perturbation, and combined optical and SAR modal perturbation. The detailed experimental results are shown in Table VI.

We can see from Table VI that the addition of the optical modal perturbation may not yield an accuracy improvement compared with the modal with no perturbation. We analyze that this is because the information contained in the optical

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TABLE VI
EFFECTS OF TUPLE PERTURBATION ON ACCURACY (IN %)

Dataset	Module	OA	Kappa
YESeg-OPT-SAR	No perturbation	79.14	64.18
	Optical modal perturbation	78.93	63.09
	SAR modal perturbation	79.43	64.39
	Combined optical and SAR modal perturbation	79.82	73.15
WHU-OPT-SAR	No perturbation	82.55	72.92
	Optical modal perturbation	82.59	73.04
	SAR modal perturbation	82.68	73.15
	Combined optical and SAR modal perturbation	82.79	73.37

modal is far more than that contained in the SAR modal, and the optical modal perturbation will make the difference between the generated negative samples and the anchor samples larger, which can be easily distinguished, and the training effect is limited. At the same time, the difference between the negative samples generated by SAR modal perturbation and the anchor samples is small and more challenging. Using such samples to train the model can significantly improve the model performance. This ablation study illustrates that the combined optical and SAR modal perturbation plays a positive role in improving the model performance. Therefore, combined optical and SAR modal perturbation is adopted in TMCNet.

2) Effectiveness of Hyperparameter γ: The hyperparameter γ in (6) is set to balance the global contrastive loss and the local contrastive loss. We believe that the global contrastive learning module and the local contrastive learning module can learn different information useful for semantic segmentation tasks, both of which are very important, so we set $\gamma = 0.5$ by default. However, this may not be optimal, so we further explore the effect of different γ in this section. The experimental results are shown in Fig. 11. From the experimental results, the optimal γ on both YESeg-OPT-SAR and WHU-OPT-SAR datasets is 0.5. In addition, the performance will drop rapidly in datasets when our GLCNet method has only a local contrastive learning module, i.e., $\gamma = 0$, or when our GLCNet method has only a global contrastive learning module, i.e., $\gamma = 1$. This ablation study shows that the combination of global and local contrastive losses plays an active role in improving the model performance. Therefore, TMCNet adopts global and local joint contrastive losses.

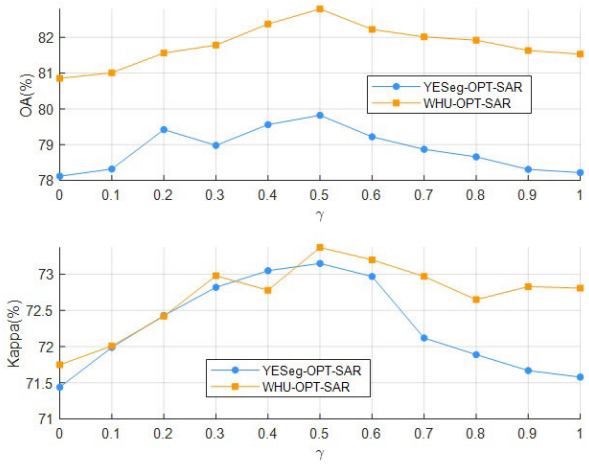


Fig. 11. Ablation study of weight γ. $\gamma = 0$ means that there is only a local matching contrastive learning module during self-supervised training, and $\gamma = 1$ means that there is only a global style contrastive learning module.

V. CONCLUSION

In this work, we integrate a contrastive learning method with the task of MRSISS, learning from an abundance of unlabeled images to extract both shared and different optical features and SAR features with the intent to lessen the dependence on labeled samples. Moreover, since existing multimodal contrastive learning methods predominantly concentrate on the stronger modality at the expense of the weaker one, these approaches could prove less optimal for MRSISS tasks that depend on multimodal features. To address this issue, we propose a new multimodal contrastive learning framework of RSISS, which consists of a TMCNet and an MSSNet. The tuple perturbation module within TMCNet makes the features of different modalities obtained by a feature extractor fused in a more compact way, which tends to lead to better semantic segmentation performance. The proposed MSSNet uses the complementary information between images to reduce noise and fuse multimodal features more closely. The experimental results show that TMCNet and MSSNet have better performance in MRSISS tasks than other state of art methods. In addition, the increase in self-supervised pretraining samples correlates with performance enhancements. Given that a large amount of remote sensing data is readily obtainable, our method harbors the potential for substantial practical utility.

In this article, we primarily conduct multimodal semantic segmentation experiments on optical and SAR images. While we have achieved notable results, the fusion of multimodal data increases model complexity and necessitates greater computational resources and storage capacity. In future work, we aim to develop a more lightweight model and investigate images with more modalities, including infrared, multispectral, and hyperspectral images.

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